CHEM 1043 Appendix
Physical Constants

Planck's constant	$h = 6.626 \times 10^{-34}$ J s molecule^{-1}
Velocity of light in vacuum	$c = 2.998 \times 10^{8}$ m s^{-1}
Avogadro's number	$N = 6.022 \times 10^{23}$ molecules mol^{-1}
Boltzmann constant	$k = 1.381 \times 10^{-23}$ J K^{-1} molecule^{-1}
Gas constant	$R = 8.314$ J K^{-1} mol^{-1} $= 0.08206$ dm^3 atm K^{-1} mol^{-1} $= 0.08314$ dm^3 bar K^{-1} mol^{-1} $= 8.314$ m^3 Pa K^{-1} mol^{-1}
Faraday constant	$F = 96485$ C mol^{-1} (of electrons transferred) $= 23061$ cal volt^{-1} mol^{-1} (of electrons transferred)
Elementary charge	$e = 1.602 \times 10^{-19}$ C
Mass of electron	$m_e = 9.109 \times 10^{-31}$ kg
Mass of proton	$m_p = 1.673 \times 10^{-27}$ kg
Atomic mass constant	$m_u = 1.661 \times 10^{-27}$ kg
Standard gravitational acceleration	$g = 9.807$ m s^{-2}

Conversion Tables:	Numerical Constants:
1 cal = 4.184 J (exactly) 1 erg = 10^{-7} J 1 G = 10^{-4} T (G = gauss, T = tesla) 1 atm = 101325 Pa = 760 mmHg (Torr) 1 bar = 10^5 Pa 0 °C = 273.15 K	π = 3.141593 e = 2.718282 ln x = 2.303 log x

Energy Conversion Factors:				
	cal mol^{-1}	J molecule^{-1}	eV molecule^{-1}	cm^{-1}
1 cal mol^{-1} =	1	6.947×10^{-24}	4.336×10^{-5}	0.3500
1 J molecule^{-1} =	1.439×10^{23}	1	6.242×10^{18}	5.036×10^{22}
1 eV molecule^{-1} =	2.306×10^{4}	1.602×10^{-19}	1	8.068×10^{3}
1 cm^{-1} =	2.859	1.986×10^{-23}	1.240×10^{-4}	1

Some Equations for Kinetic Theory of Gases and Electrochemistry:

$u_{rms} = \sqrt{\frac{3RT}{M}}$	$\overline{E_k} = \frac{3}{2}\left(\frac{RT}{N_A}\right)$	$P = \frac{1}{3}\frac{N}{V}m\overline{u^2}$
$\Delta_r G^\circ = -zFE^\circ_{cell}$	$E^\circ_{cell} = \frac{0.025693\ V}{z}\ln K$	Nernst equation: $E_{cell} = E^\circ_{cell} - \frac{0.0257\ V}{z}\ln Q$